

IED 1.8 | Engineering Documentation Challenge

Unit 1 Final Challenge | Student Packet



PAGE 1

NAME: _____ PERIOD: _____ DATE: _____

MISSION BRIEF

Use evidence, not guessing

You will document a small purpose-made assembly so another engineering team could understand, reproduce, inspect, and improve it. The physical assembly is your source of evidence. Your final package should communicate the design clearly without requiring a verbal explanation.

DRIVING QUESTION

How can an engineer fully communicate an existing assembly so another person could reproduce it without guessing?

FINAL DELIVERABLE

- isometric overview sketch
- aligned front/top/right-side views
- complete dimensions
- measurement and tolerance evidence
- joining/interface analysis
- justified improvement idea
- peer review and final revision check

CHALLENGE RULES

Keep your work neat enough for another team to use

1. Measure the actual assembly before finalizing critical dimensions.
2. Do not dimension from memory or guess at hidden features.
3. Use proper line conventions, centerlines, hidden lines, and dimension placement.
4. Avoid repeated dimensions. Every important feature should be sized and located.
5. Use tolerances only where they help control function, fit, alignment, or assembly.
6. Your improvement must be based on evidence from the existing design.

ASSEMBLY SNAPSHOT

Complete this before detailed documentation

Assembly name or station ID: _____

Primary function of the assembly: _____

Main parts you can identify: _____

Most important interface or connection: _____

IED 1.8 | Product Investigation

Observe the assembly before drawing it

NAME: _____

PERIOD: _____

DATE: _____

PARTS AND INTERFACES

Identify how the assembly is built

Part ID	Part name/description	Material/process	Function in assembly	Qty
A				
B				
C				
D				

INTERFACE MAP

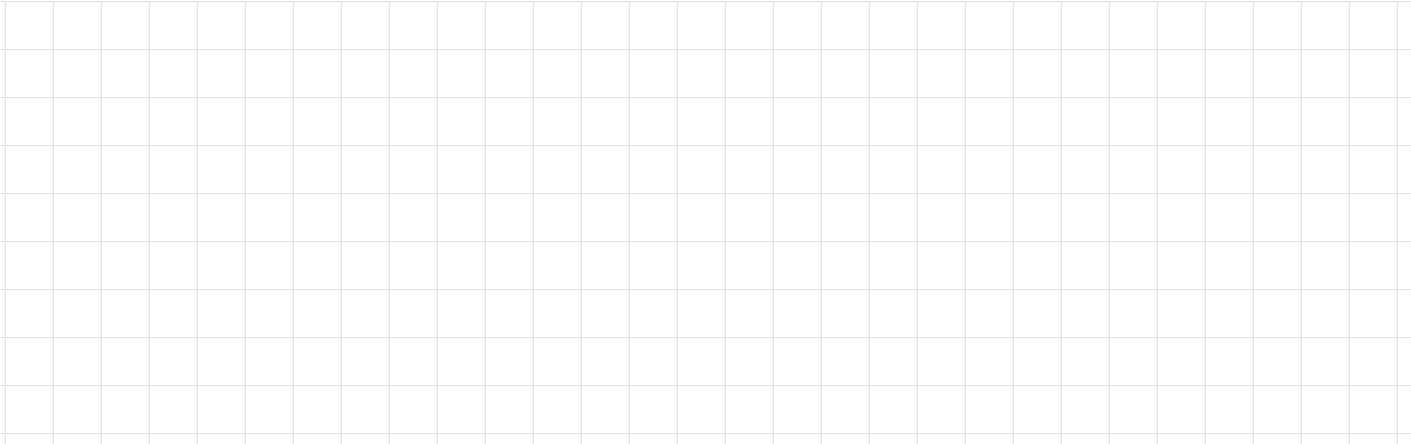
Where do parts touch, connect, align, or move?

Interface	Parts	Connection	Function	Evidence
1				
2				
3				
4				

ISOMETRIC OVERVIEW SKETCH

Show the full assembly in one clear view

ISOMETRIC SKETCH SPACE



Label major parts and important interfaces.

IED 1.8 | Multiview Documentation

Create aligned views using proper line conventions

NAME: _____ PERIOD: _____ DATE: _____

VIEW PLANNING

Choose views that best communicate the assembly

Front view chosen because: _____

Top view should show: _____

Right-side view should show: _____

Hidden or center lines needed for: _____

MULTIVIEW SKETCH SPACE

Align Front, Top, and Right-side views

FRONT VIEW



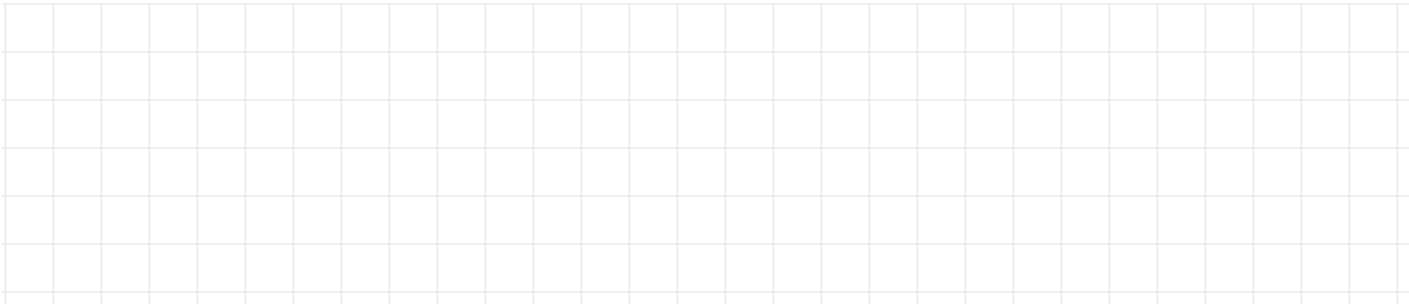
Use object, hidden, and center lines.

RIGHT-SIDE VIEW



Keep views aligned where possible.

TOP VIEW



Line up features with the front view.

IED 1.8 | Measurement and Dimensioning

Size and locate the features another team needs

NAME: _____ PERIOD: _____ DATE: _____

MEASUREMENT RECORD

Measure before final dimension decisions

Feature measured	Tool used	Trial 1	Trial 2	Trial 3	Reported value

CRITICAL DIMENSIONS AND TOLERANCES

Choose dimensions that control function or fit

Critical feature	Nominal value	Tolerance	Why it matters	What happens if wrong?

Dimensioning check: Every needed feature is sized and located. No dimensions are repeated unnecessarily.

IED 1.8 | Interface and Joining Analysis

Explain how the assembly works, not just what it looks like

NAME: _____ PERIOD: _____ DATE: _____

JOINING / INTERFACE ANALYSIS

Use the decision factors from Lesson 1.7

Connection	Method	Removable?	Function	Best reason	Concern

ENGINEERING IMPROVEMENT PROPOSAL

Base your idea on evidence from the existing assembly

Problem or opportunity observed: _____

Evidence from measurements, sketches, or interface analysis: _____

Proposed design change: _____

How the change improves function, assembly, serviceability, or manufacturability: _____

Sketch of proposed improvement:

IMPROVEMENT SKETCH



Label what changed.

IED 1.8 | Peer Review and Final Check

Could another team reproduce this without guessing?

NAME: _____ PERIOD: _____ DATE: _____

PEER ENGINEERING REVIEW

Reviewer completes this section

1. The assembly purpose is clear. ☐
2. Major parts are identified and labeled. ☐
3. Views are aligned and use correct line types. ☐
4. Features are sized and located without obvious missing dimensions. ☐
5. Critical dimensions and tolerances are reasonable. ☐
6. Joining/interface analysis explains how the assembly works. ☐
7. The improvement proposal is supported by evidence. ☐

Reviewer name: _____

One strength of this documentation package: _____

One specific revision needed before final submission: _____

FINAL SUBMISSION RUBRIC

Use this before turning in your package

Category	Points	What strong work includes
Sketching and views	20	Clear isometric sketch; aligned orthographic views; correct line conventions
Dimensioning	20	Complete sizes and locations; proper symbols; no repeated or missing key dimensions
Measurement and tolerances	20	Accurate records; critical dimensions; tolerances tied to function or fit
Joining/interface analysis	15	Connection method, parts involved, function, and tradeoffs explained
Improvement proposal	15	Evidence-based change with clear sketch and justification
Professional quality	10	Neat, organized, labeled, readable, and complete

Final revision completed by: _____